NABIEL MUHAMMAD IMJAUZANANSYAH

Banda Aceh, Indonesia | +62 852-1147-9415 | nabiel.zmuhammad11@gmail.com | github.com/NabielMuhammad | linkedin.com/in/nabiel-muhammad

PROFESSIONAL SUMMARY

Fresh Information Systems graduate (GPA 3.66/4.00) specializing in applied machine learning and AI systems. Built and deployed a CNN-LSTM computer vision model achieving 96% validation accuracy for real-time sign language translation, published in Jurnal ICT. Hands-on experience with LSTM time-series forecasting, LLM/agentic integration (Google Gemini API), and NLP classification through independent projects and the DBS Foundation Machine Learning Engineering program. Comfortable working end-to-end: data collection, model training and evaluation, and production deployment behind REST APIs.

EDUCATION

Bachelor of Information Systems, Universitas Muhammadiyah Sumatera Utara (UMSU)

Medan, Indonesia | 2022 - June 2026

- GPA: 3.66 / 4.00
- Final year project: real-time BISINDO sign language translator using CNN-LSTM and Text-to-Speech, achieving 96% validation accuracy
- published in Jurnal ICT: Information and Communication Technologies (2026)
- Relevant coursework: Machine Learning, Data Analysis, Database Systems, Algorithm and Programming, Software Engineering, Web Development

RELEVANT EXPERIENCE

Machine Learning Engineer Trainee, DBS Foundation Coding Camp 2025

Remote, Part-time | Feb 2025 - Aug 2025

- Completed an intensive machine learning engineering program covering data preprocessing, model development, evaluation, and deployment workflows
- Built and evaluated data-driven prototypes, including a recommendation system and an NLP-based text classification model, using Python, Scikit-learn, TensorFlow, and Keras
- Practiced reproducible project structuring, technical documentation, and deployment preparation to support production-ready ML workflows

Youth Sales Strategist Intern, PT Telekomunikasi Selular

Medan, Indonesia | Jun 2025 - Sep 2025

- Developed a Python Telegram voucher bot for BGA SUMBAGUT (program, challenge, survey, point redemption, and profile menus), integrated with the SAMOSIR.LIVE web platform via API
- Built SAMOSIR web modules in Laravel (survey and challenge management, wallet and redeem, broadcast messaging, Telegram activity log, CSV export), and migrated sales data from Excel into the database managed with Navicat
- Implemented the points system with two-step verification for redeem transactions, fallback handling for invalid or duplicate input, a cluster-based redeem approval flow, and a user whitelist
- Deployed the system to the BGA internal server, fixed post-deployment bugs, and wrote technical documentation for the points flow, API integration, and deployment procedure
- Completed 100% of the planned Web and Bot development scope (target: 80%), scoring 98/100 on the performance target in the mentor's evaluation

PROJECTS

NzSignify - Real-Time BISINDO Sign Language Translator

Final Year Project | Dec 2025 - Apr 2026

- Developed a real-time Indonesian Sign Language (BISINDO) translation system using a hybrid CNN-LSTM deep learning architecture for gesture classification
- Collected and preprocessed an original dataset of 11,000 image frames across 11 gesture classes from 3 participants, applying spatio-temporal normalization and 30-frame sequence extraction
- Trained and optimized the model in Google Colab, achieving 96% classification accuracy, with confidence-threshold filtering to improve prediction reliability

- Built a React.js frontend integrated with a Node.js backend to process real-time webcam streams and communicate with the AI inference service via REST API
- Implemented Text-to-Speech (TTS) functionality to convert recognized gestures into spoken output, improving accessibility for end users

InvestMentor - AI-Powered Trading Simulator & Agentic Mentor Platform

Full-Stack AI Engineer | Independent Project | Sep 2025 - Jun 2026

- Engineered an interactive AI investment mentor using the Google Gemini 2.0 Flash SDK, implementing multi-turn conversational memory and context-aware responses for stock market analysis
- Deployed an LSTM-based machine learning model using TensorFlow.js to generate daily stock price predictions with calculated confidence levels
- Built real-time data ingestion pipelines integrating the Yahoo Finance API and Google News RSS, serving dynamic market context to the frontend via WebSocket (Socket.IO)
- Developed a Node.js/Express.js backend with a secure REST API architecture, implementing API rate limiting, JWT authentication, and database transactions via Sequelize ORM

Additional projects: ParentCare - AI-powered parenting style platform (Random Forest classifier, team of 6, DBS Foundation Capstone, 2025); Fashion Studio ETL Pipeline - independent Python data pipeline (Mar 2025).

TECHNICAL SKILLS

AI & Machine Learning: TensorFlow, TensorFlow.js, Keras, Scikit-learn, CNN-LSTM, LSTM, NLP, Computer Vision, LLM/Generative AI Integration (Google Gemini API), Model Evaluation & Deployment

Programming Languages: Python, JavaScript, Node.js, PHP

Data Engineering & Analytics: ETL Pipelines, Real-Time Data Pipelines, Pandas, NumPy, Data Cleaning & Validation, REST API Integration, WebSocket (Socket.IO), Google Sheets API

Backend & Database: Express.js, Laravel, JWT Authentication, Sequelize ORM, MySQL, PostgreSQL, SQL Fundamentals, Relational Database Design

Tools & Platforms: Git, GitHub, Docker, Google Colab, VS Code, Vercel, Railway

CERTIFICATIONS

- DBS Foundation Coding Camp: Machine Learning Engineering - Dicoding Indonesia (2025)
- Applied Machine Learning, Machine Learning Development, Machine Learning for Beginners - Dicoding & Google Developers (2025)
- Data Analysis with Python - Dicoding & Google Developers (2025)
- Certificate of Completion: Telkomsel Internship Partnership Program 2025 - PT Telekomunikasi Selular (2025)
- Additional: SQL Fundamentals, Git and GitHub Fundamentals, Fundamentals of Data Processing/ETL Pipelines - Dicoding & Google Developers (2025)